

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and state it clearly in the answer-sheet.

	(CO)	Marks
Q. 1 Solve any two of the following.		12
A) What is the principle of stabilization pond? Give Construction details for Oxidation pond with neat and clean diagram.	CO1	6
B) What is Self purification of natural streams. Also explain various actions involved in Self purification.	CO1	6
C) Design a septic tank for the following data: No. of people = 100 Sewage/capita/day = 120 litres De-sludging period = 1 year Length : width = 4:1 What would be the size of its soak well if the effluent from this septic tank is to be discharged in it. Assume percolation rate through the soak well to be $1250 \text{ l/m}^2/\text{d}$.	CO1	6
Q.2 Solve any two of the following.		12
A) What is industrial pollution? Discuss control measures for industrial pollution.	CO2	6
B) State the objectives and advantages of equalization. Explain briefly Equalization process in industrial waste water treatment.	CO2	6
C) What is neutralization? Explain any two methods of neutralization in industrial waste water treatment.	CO2	6
Q. 3 Solve any two of the following.		12
A) Explain Nitrification and Denitrification process in industrial waste water treatment.	CO3	6
B) What do you mean by CETP ? Explain the need of CETP.	CO3	6
C) Explain Membrane filtration process with neat and clean diagram.	CO3	6

Q.4	Solve any two of the following.		12
A)	What do you mean by communicable diseases? Also explain 5-F diagram with respect to Faecal- oral transmitted diseases.	CO4	6
B)	Explain following diseases with respect to infections agent, occurrence, reservoir, mode of transmission and control. (i) Cholera (ii) Tuberculosis	CO4	6
C)	Explain Chain of Disease Transmission.	CO4	6
Q. 5	Solve any two of the following.		12
A)	List out various low cost excreta disposal methods. Explain Aqua Privy and Borehole Privy.	CO4	6
B)	Explain Rural Sanitation in details.	CO4	6
C)	Explain Life cycle of Mosquito. Also enlist various vector borne diseases and explain any one in brief.	CO4	6

***** End *****